Week_6_GIT/Exercise_4_Merge_Conflict_Resolution

Step 1: Open Git Bash

Launch **Git Bash**.

Step 2: Navigate to Your Repository

cd ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling

Verify that the repository is clean.

git status

Expected Output

On branch main

nothing to commit, working tree clean

CTS first requires verifying that the master (main) branch is in a clean state.

Part 1 – Create a Branch

Step 3: Create a New Branch

Create a branch named **GitWork**.

git branch GitWork

Step 4: Switch to the Branch

git checkout GitWork

or

git switch GitWork

Expected Output

Switched to branch 'GitWork'

Step 5: Create hello.xml

echo "<message>Hello from GitWork</message>" > hello.xml

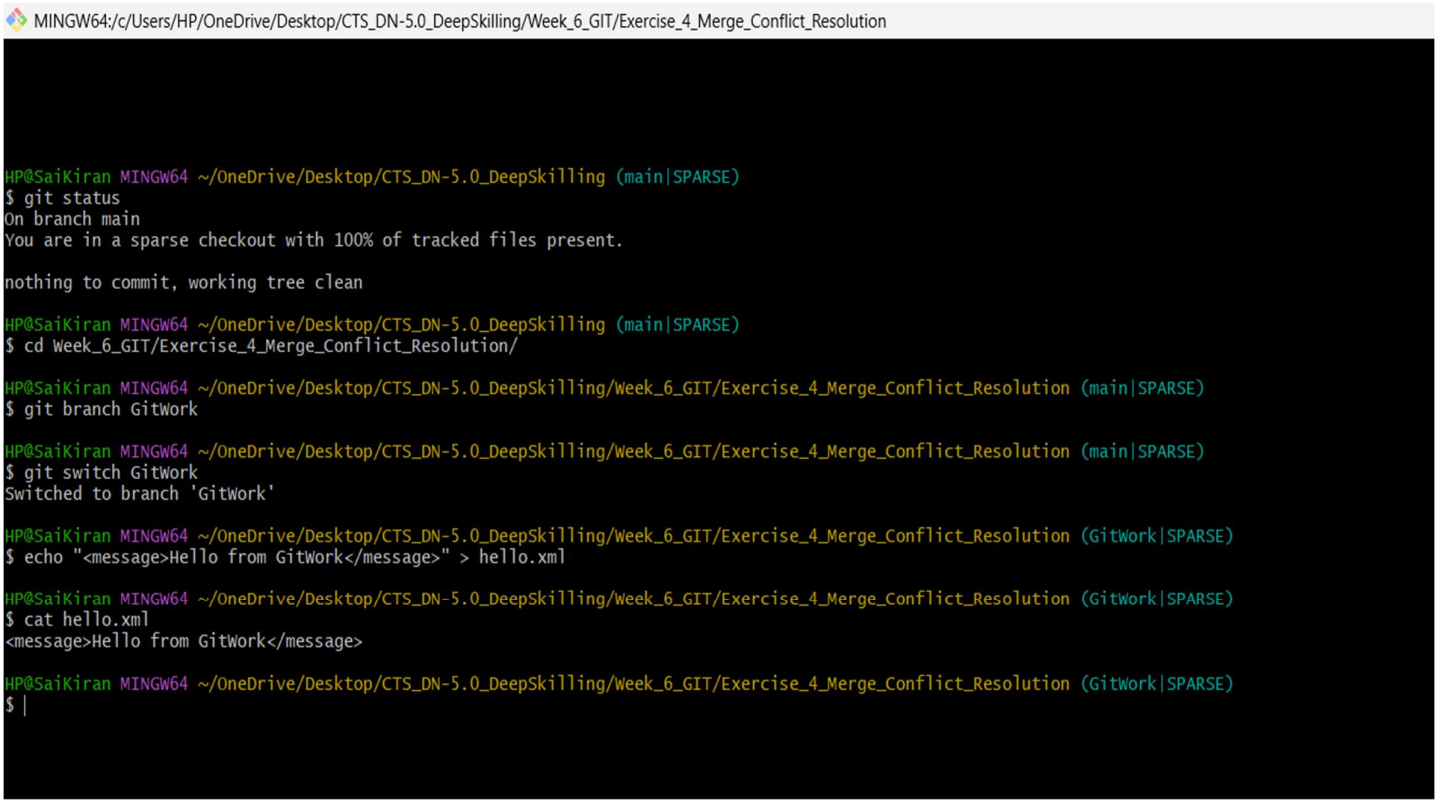
Verify the file.

cat hello.xml

Expected Output

<message>Hello from GitWork</message>

CTS specifies creating a file named **hello.xml** in the branch.



Step 6: Modify hello.xml

Replace the contents.

echo "<message>Updated from GitWork Branch</message>" > hello.xml

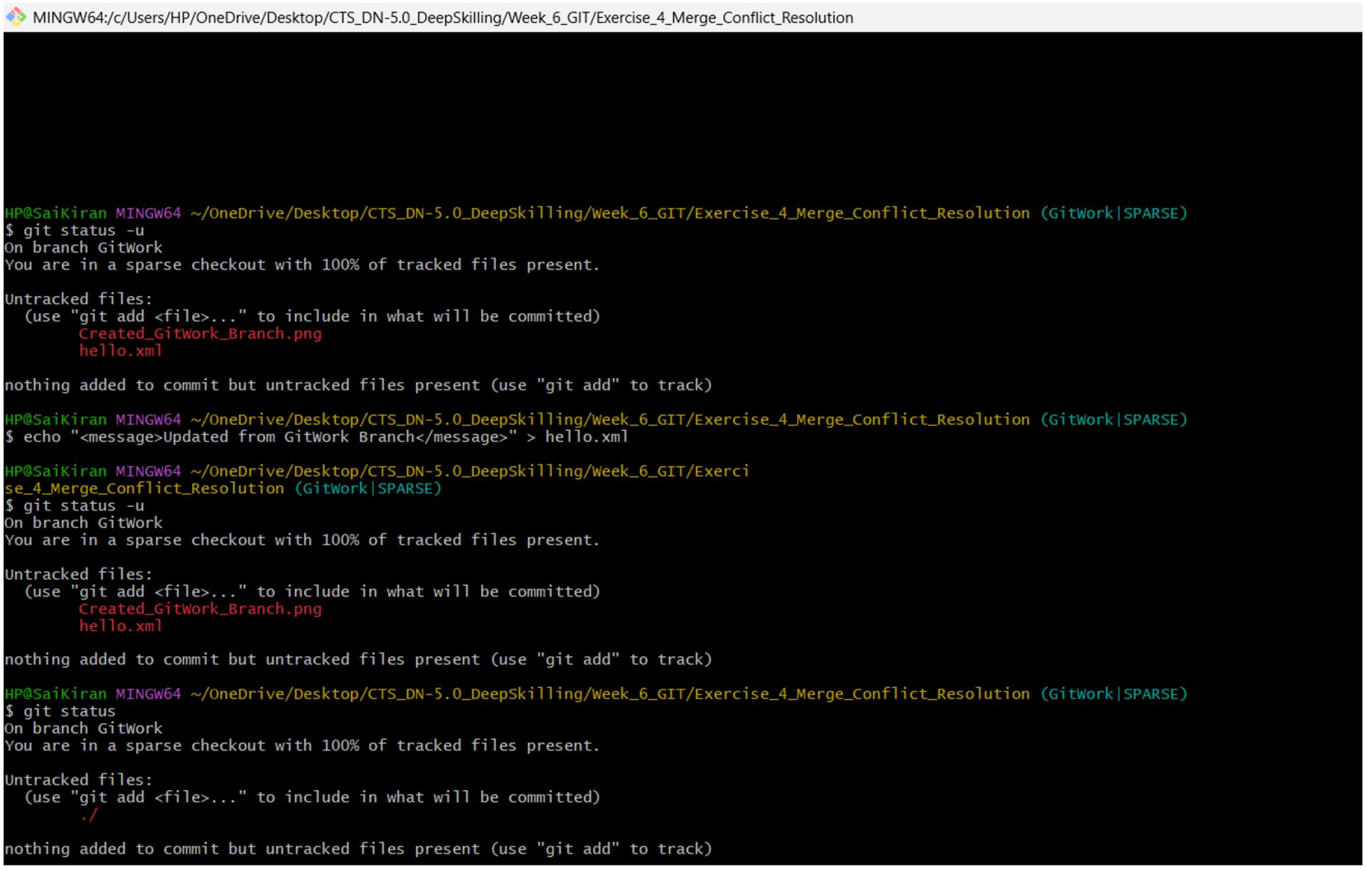
Check repository status.

git status

Expected Output

modified: hello.xml

CTS requires updating the file and observing the status.



Step 7: Commit the Branch Changes

git add hello.xml

git commit -m "Update hello.xml in GitWork branch"

Expected Output

[GitWork xxxxxx]

Update hello.xml in GitWork branch

Part 2 – Modify Main Branch

Step 8: Switch Back to Main

git checkout main

or

git switch main

Step 9: Create the Same File with Different Content

echo "<message>Hello from Main Branch</message>" > hello.xml

Now overwrite it with different content.

echo "<message>Updated from Main Branch</message>" > hello.xml

Verify:

cat hello.xml

Expected Output

<message>Updated from Main Branch</message>

The CTS exercise instructs adding the same file with different content in the master branch to intentionally create a conflict.

Step 10: Commit the Main Branch Changes

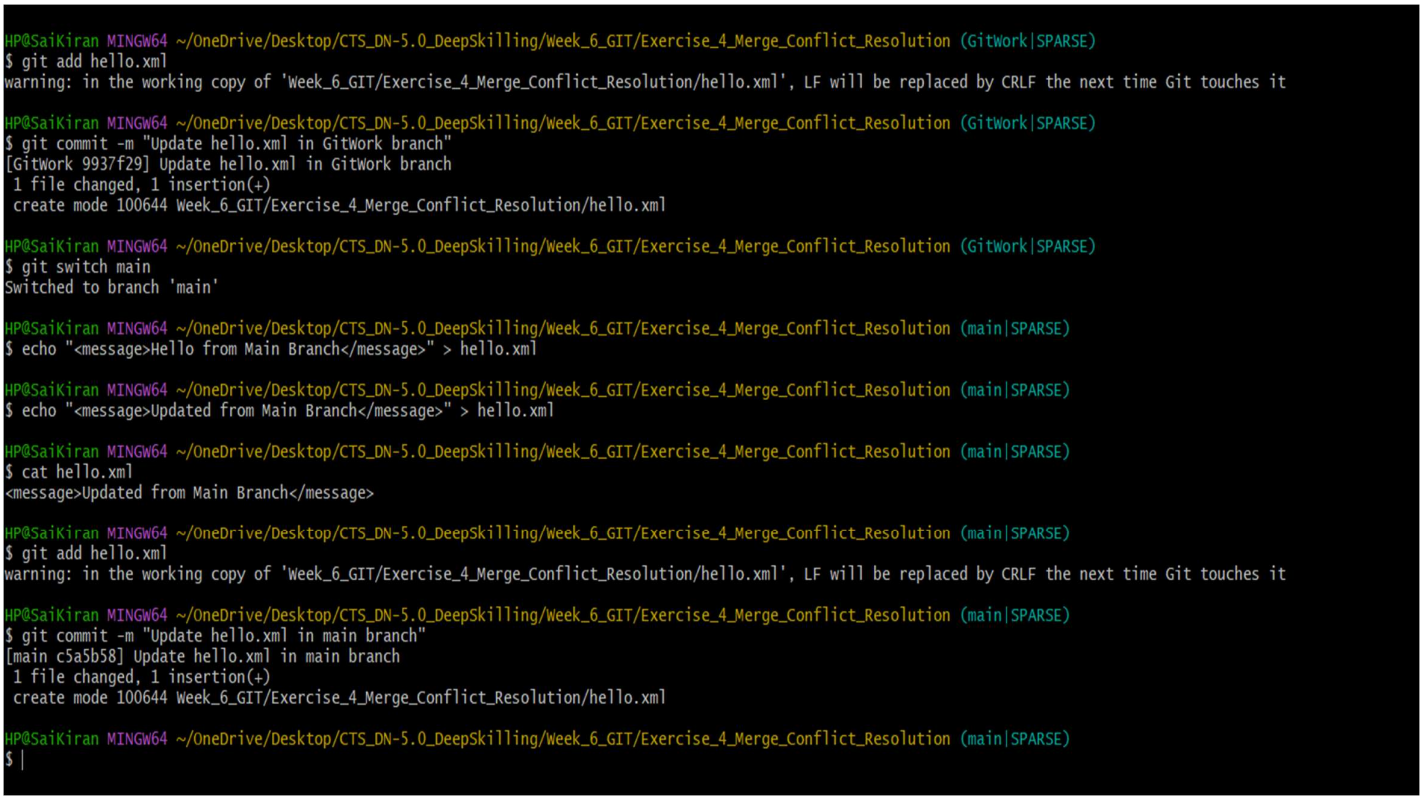
git add hello.xml

git commit -m "Update hello.xml in main branch"

Expected Output

[main xxxxxx]

Update hello.xml in main branch



Part 3 – View Branch History

Step 11: Display the Commit Graph

git log --oneline --graph --decorate --all

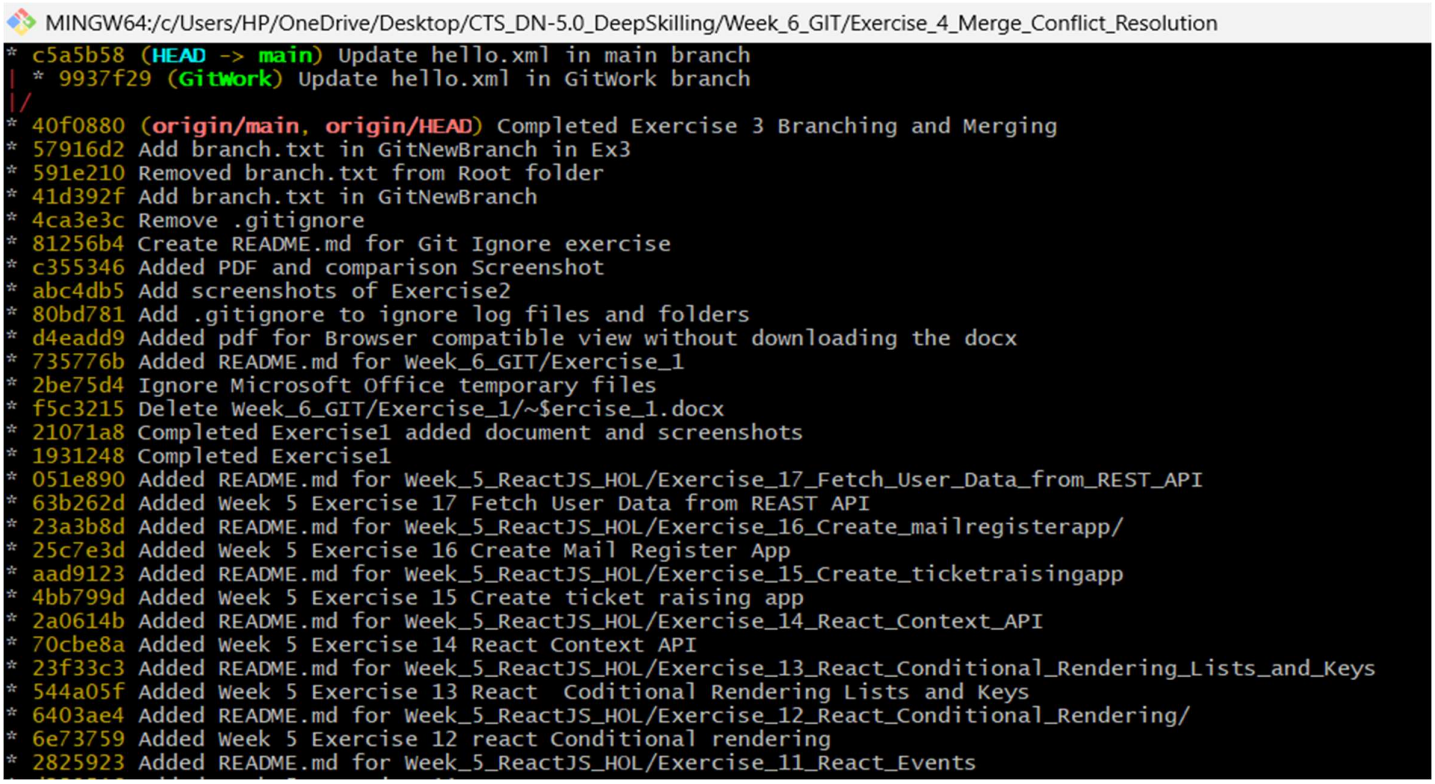
Example Output

* abc1234 (HEAD -> main)

* def5678 (GitWork)

* ghi9012 Initial Commit

CTS instructs viewing the complete commit graph before merging.



Part 4 – Compare the Branches

Step 12: Compare Using Git Diff

git diff main GitWork

Git displays the differences between the two branches.

CTS requires checking the differences using the Git diff tool.

Step 13: Compare Using P4Merge (Optional)

If P4Merge is configured:

git difftool main GitWork

This opens a graphical comparison of the conflicting files.

CTS recommends P4Merge for better visualization.

Part 5 – Create the Merge Conflict

Step 14: Merge GitWork into Main

git merge GitWork

Expected Output

Auto-merging hello.xml

CONFLICT (content): Merge conflict in hello.xml

Automatic merge failed; fix conflicts and then commit the result.

This conflict occurs because both branches modified the same file differently, which is the intended outcome of this exercise.

Step 15: Observe the Conflict Markers

Open the file:

cat hello.xml

Git inserts conflict markers similar to:

<<<<<<< HEAD

<message>Updated from Main Branch</message>

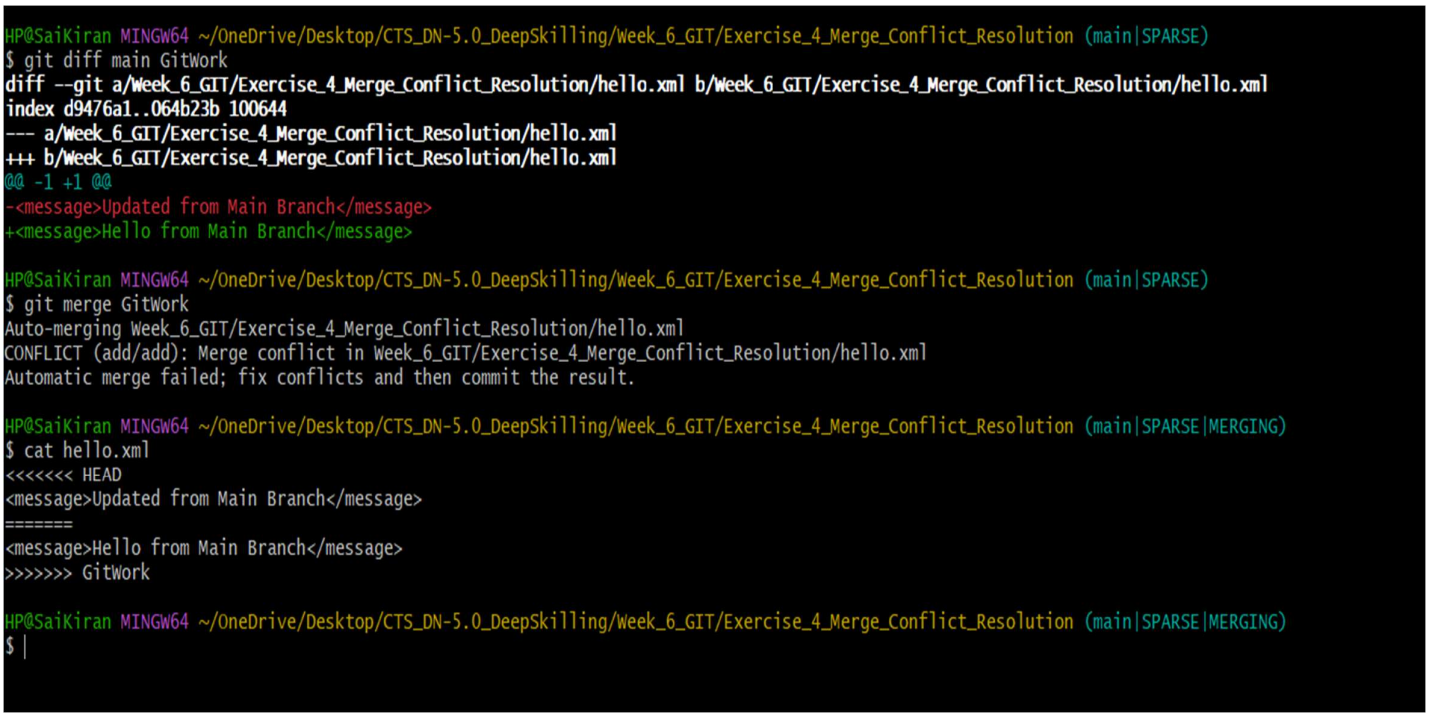
=======

<message>Updated from GitWork Branch</message>

>>>>>>> GitWork

These markers indicate the conflicting changes.

CTS refers to observing the Git markup after merging.



Part 6 – Resolve the Conflict

Step 16: Edit hello.xml

Remove the conflict markers and keep the desired content.

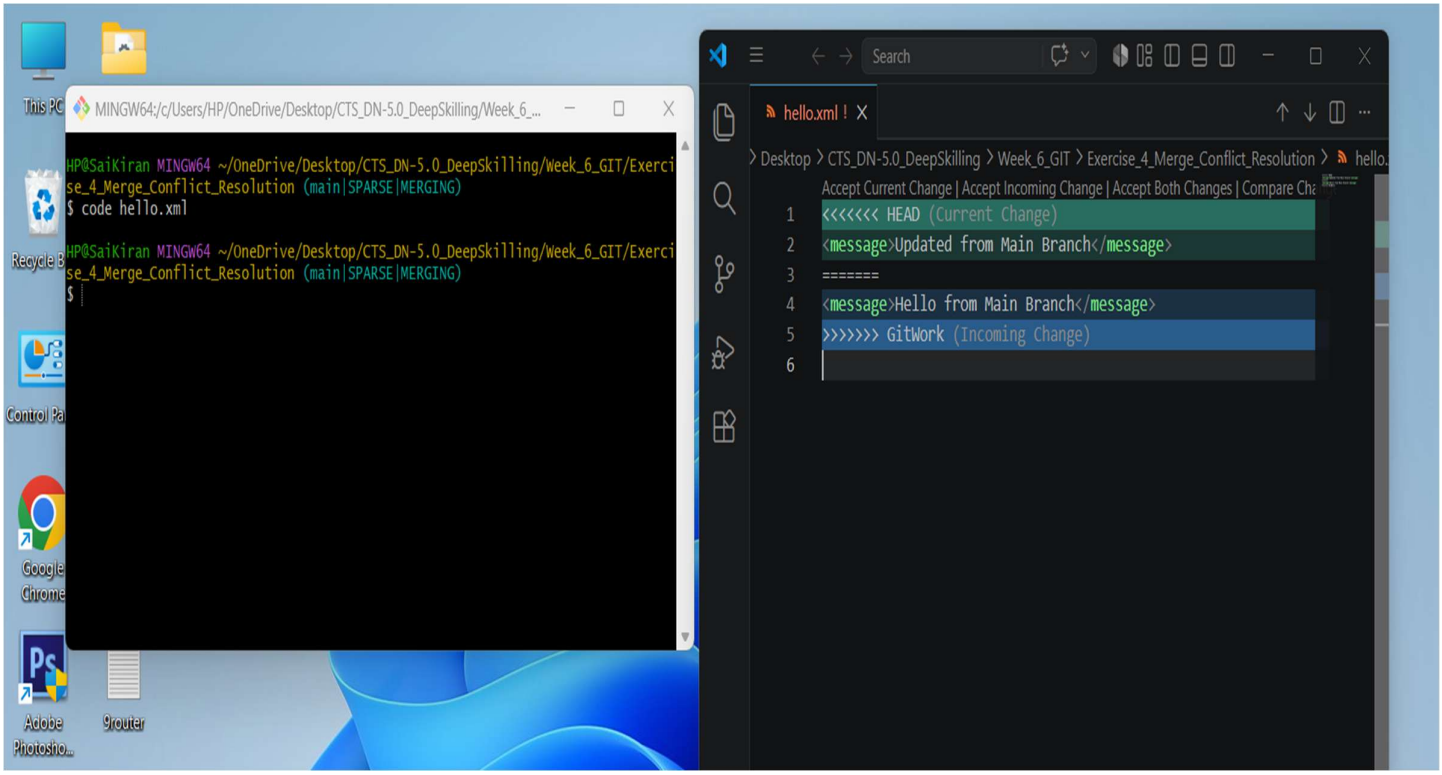
Example:

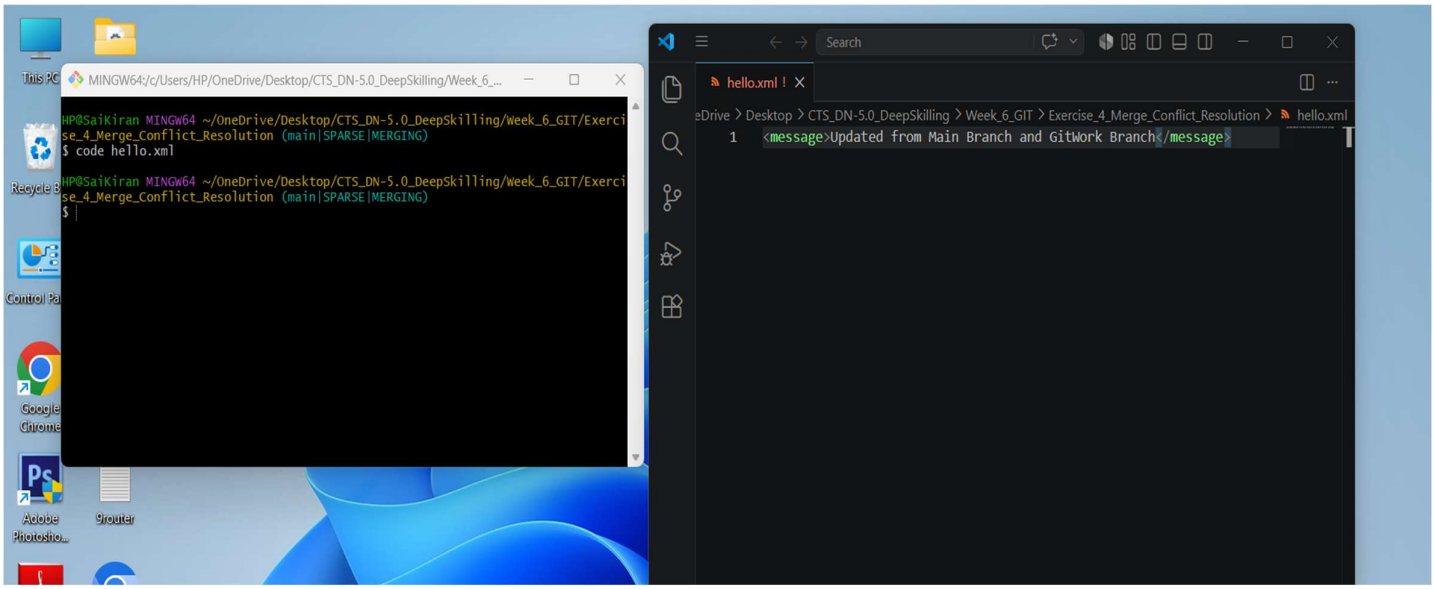
<message>Updated from Main Branch and GitWork Branch</message>

Save the file.

If using **P4Merge**, resolve the conflict using its 3-way merge interface.

CTS specifically requires using a 3-way merge tool to resolve the conflict.





Step 17: Stage the Resolved File

git add hello.xml

Step 18: Complete the Merge

git commit -m "Resolve merge conflict in hello.xml"

Expected Output

[main xxxxxx]

Resolve merge conflict in hello.xml

CTS instructs committing the resolved merge.

Part 7 – Ignore Backup Files

Step 19: Check Repository Status

git status

If your merge tool created backup files (for example, hello.xml.orig), add them to .gitignore.

Open or create .gitignore and add:

*.orig

Stage the .gitignore file:

git add .gitignore

Commit:

git commit -m "Ignore backup files"

CTS requires adding backup files to .gitignore and committing the changes.

Part 8 – Clean Up

Step 20: List All Branches

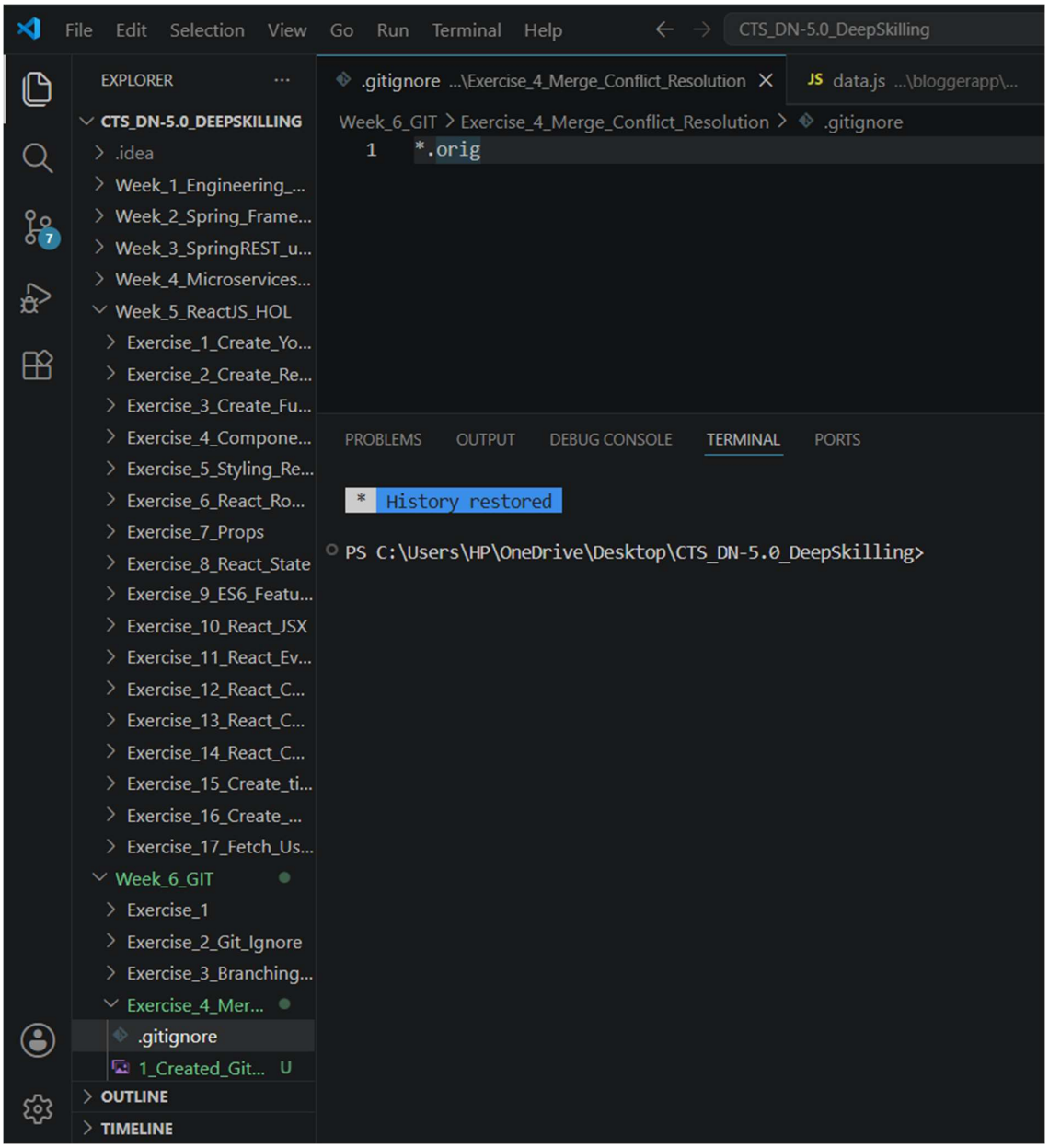
git branch

Example Output

GitWork

* main

```
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE|MERGING)
$ git add hello.xml

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE|MERGING)
$ git commit -m "Resolve merge conflict in hello.xml"
[main 48d5447] Resolve merge conflict in hello.xml

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git status
On branch main
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        1_Created_GitWork_Branch.png
        2_Updated_hello.xml.png
        3_Committed_hello.xml_in_main_branch.png
        4_git_log.png
        5_Merge_conflict.png
        6_Editing_in_VSC.png
        8_Editted_hello.xml.png

nothing added to commit but untracked files present (use "git add" to track)

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git add .gitignore

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git commit -m "Ignore backup files"
[main 39f7f9c] Ignore backup files
 1 file changed, 1 insertion(+)
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/.gitignore

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git branch
  GitWork
* main

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$
```



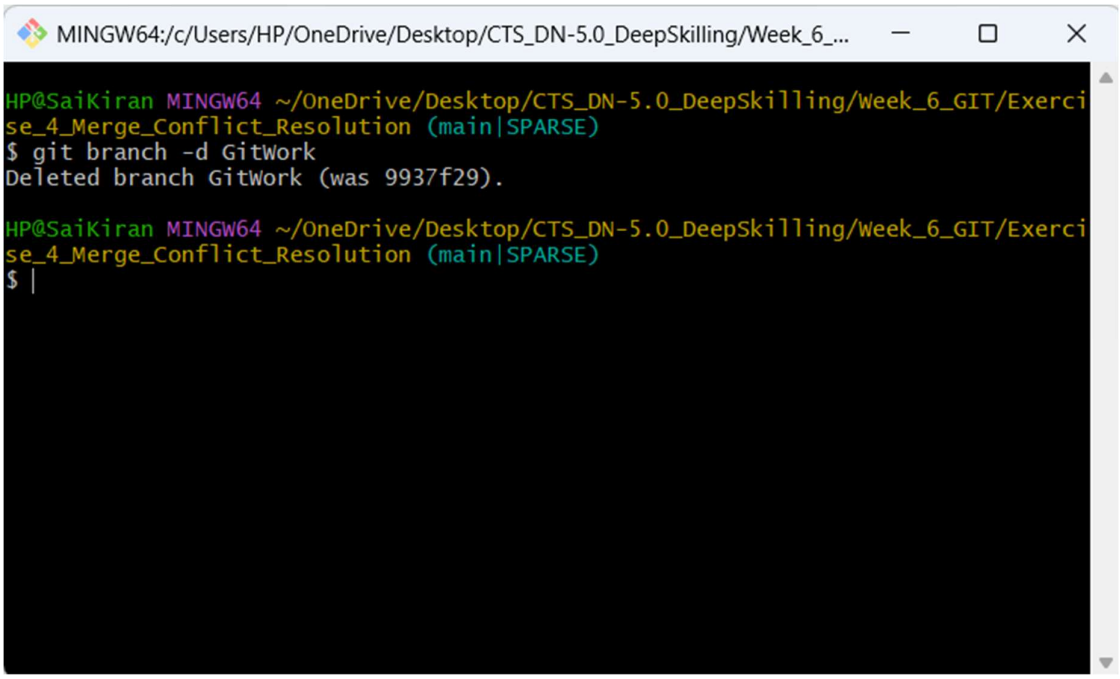
Step 21: Delete the Merged Branch

git branch -d GitWork

Expected Output

Deleted branch GitWork (was abc1234).

CTS instructs deleting the merged branch.



Step 22: View the Final Commit Graph

git log --oneline --graph --decorate

Example Output

* 9ab4567 (HEAD -> main) Resolve merge conflict in hello.xml

* 8cd3456 Merge branch 'GitWork'

* 7ef2345 Update hello.xml in main branch

* 6de1234 Update hello.xml in GitWork branch

* 5bc0123 Initial Commit

This confirms the merge conflict was resolved successfully and the history reflects the merge.

```
MINGW64:/c/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution
* 39f7f9c (HEAD -> main) Ignore backup files
*   48d5447 Resolve merge conflict in hello.xml
|\
| * 9937f29 Update hello.xml in GitWork branch
* | c5a5b58 Update hello.xml in main branch
|/
* 40f0880 (origin/main, origin/HEAD) Completed Exercise 3 Branching and Merging
* 57916d2 Add branch.txt in GitNewBranch in Ex3
* 591e210 Removed branch.txt from Root folder
* 41d392f Add branch.txt in GitNewBranch
* 4ca3e3c Remove .gitignore
* 81256b4 Create README.md for Git Ignore exercise
* c355346 Added PDF and comparison Screenshot
* abc4db5 Add screenshots of Exercise2
* 80bd781 Add .gitignore to ignore log files and folders
* d4eadd9 Added pdf for Browser compatible view without downloading the docx
* 735776b Added README.md for Week_6_GIT/Exercise_1
* 2be75d4 Ignore Microsoft Office temporary files
* f5c3215 Delete Week_6_GIT/Exercise_1/~$ercise_1.docx
* 21071a8 Completed Exercise1 added document and screenshots
* 1931248 Completed Exercise1
* 051e890 Added README.md for Week_5_ReactJS_HOL/Exercise_17_Fetch_User_Data_from_REST_API
:...skipping...
* 39f7f9c (HEAD -> main) Ignore backup files
*   48d5447 Resolve merge conflict in hello.xml
|\
| * 9937f29 Update hello.xml in GitWork branch
* | c5a5b58 Update hello.xml in main branch
|/
* 40f0880 (origin/main, origin/HEAD) Completed Exercise 3 Branching and Merging
* 57916d2 Add branch.txt in GitNewBranch in Ex3
* 591e210 Removed branch.txt from Root folder
* 41d392f Add branch.txt in GitNewBranch
* 4ca3e3c Remove .gitignore
* 81256b4 Create README.md for Git Ignore exercise
* c355346 Added PDF and comparison Screenshot
* abc4db5 Add screenshots of Exercise2
* 80bd781 Add .gitignore to ignore log files and folders
* d4eadd9 Added pdf for Browser compatible view without downloading the docx
```

Push the Changes to GitHub

git pull origin main

git push origin main

```
MINGW64:/c/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git add .

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git commit -m "Adding Screenshots to the Exercise 4"
[main 136ccf5] Adding Screenshots to the Exercise 4
 10 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/10_deleted_GitWork_branch.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/11_Git_log.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/1_Created_GitWork_Branch.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/2_Updated_hello.xml.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/3_Committed_hello.xml_in_main_branch.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/4_git_log.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/5_Merge_conflict.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/6_Editing_in_VSC.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/8_Editted_hello.xml.png
 create mode 100644 Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/9_Resolved_Conflict.png

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git pull origin main
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 8.00 KiB | 431.00 KiB/s, done.
From https://github.com/G-Manikanta1729/CTS_DN-5.0_DeepSkilling
 * branch            main       -> FETCH_HEAD
   40f0880..ae2185c  main       -> origin/main
Merge made by the 'ort' strategy.
 .../Exercise_3_Branching_and_Merging/README.md     | 1492 ++++++++++++++++++++
 1 file changed, 1492 insertions(+)
 create mode 100644 Week_6_GIT/Exercise_3_Branching_and_Merging/README.md
git pgit pgit p
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE)
$ git pull origin main --rebase
From https://github.com/G-Manikanta1729/CTS_DN-5.0_DeepSkilling
 * branch            main       -> FETCH_HEAD
Deletion of directory 'Week_6_GIT/Exercise_4_Merge_Conflict_Resolution' failed. Should I try again? (y/n) n
Auto-merging Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/hello.xml
CONFLICT (add/add): Merge conflict in Week_6_GIT/Exercise_4_Merge_Conflict_Resolution/hello.xml
error: could not apply 9937f29... Update hello.xml in GitWork branch
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 9937f29... # Update hello.xml in GitWork branch

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling/Week_6_GIT/Exercise_4_Merge_Conflict_Resolution (main|SPARSE|REBASE 2/4)
$ git push origin main
Enumerating objects: 41, done.
Counting objects: 100% (41/41), done.
Delta compression using up to 12 threads
Compressing objects: 100% (31/31), done.
Writing objects: 100% (37/37), 980.13 KiB | 31.62 MiB/s, done.
Total 37 (delta 13), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (13/13), completed with 2 local objects.
```

Final Project Structure

CTS_DN-5.0_DeepSkilling/

```
│
├── Week_1_Engineering_Concepts/
├── Week_2_Spring_Framework/
├── Week_3_Spring_REST/
├── Week_4_Microservices/
├── Week_5_ReactJS_HOL/
└── Week_6_GIT/
    └── Exercise_4_Merge_Conflict_Resolution/
        ├── hello.xml
        └── .gitignore
```

Command Summary

```
cd ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkilling

git status

git branch GitWork

git checkout GitWork

echo "<message>Hello from GitWork</message>" > hello.xml

echo "<message>Updated from GitWork Branch</message>" > hello.xml

git add hello.xml

git commit -m "Update hello.xml in GitWork branch"

git checkout main
```

```
echo "<message>Updated from Main Branch</message>" > hello.xml

git add hello.xml

git commit -m "Update hello.xml in main branch"

git log --oneline --graph --decorate --all

git diff main GitWork

git difftool main GitWork      # Optional

git merge GitWork

# Resolve conflict in hello.xml

git add hello.xml

git commit -m "Resolve merge conflict in hello.xml"

git status

git add .gitignore

git commit -m "Ignore backup files"

git branch

git branch -d GitWork

git log --oneline --graph --decorate
```

```
git pull origin main
```

```
git push origin main
```

This implementation follows the CTS Git Hands-on sequence: verify a clean repository, create a branch, make conflicting changes in both branches, observe the conflict, resolve it using a merge tool or manually, ignore backup files, delete the merged branch, and inspect the final commit history.